

Andrea González

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EDUCATION

New York University

Bachelor of Science in Computer Engineering

May 2026

EXPERIENCE

Medtronic

Software Engineering Intern

Jun 2026 - Aug 2026

Mounds View, MN

- Developed an AI agent that matches complaints to hazard codes cutting matching time by ~95%
- Enabled full traceability from complaints to hazard codes, strengthening risk defensibility for regulatory audits
- Built a 40-code hazard library from 400+ hazard logs, validated with reliability and post-market engineers
- Utilized agent output to identify hazard code gaps and to proactively generate missing hazard codes

NASA Lunabotics

Lead Software Engineer

Jun 2025 – May 2026

New York, NY

- Built a rover simulation in Gazebo to rigorously test algorithms that reduced physical testing time by ~40%
- Managed sprint planning and backlog via GitHub Projects using Agile methodology, coordinating work across team
- Designed a multi-node ROS architecture over TCPROS/UDPROS Wi-Fi, enabling remote perception and control

NYU Financial Operations & Treasury Division

E-commerce Assistant

Apr 2024 – May 2025

New York, NY

- Built 30+ eCommerce UStores with TouchNet to improve UX and purchasing flows for campus organizations
- Validated test environments with clients and managed production deployments, ensuring reliable launches

PROJECTS

Rover Autonomous Navigation — ROS2, Python, Ubuntu, Gazebo, RTAB-Map, Nav2

- Integrated sensor data from rotary encoders, load cells, and camera feed to develop state machines
- Built real-time 3D terrain maps using RTAB-Map, enabling reliable obstacle detection on unstructured terrain
- Deployed Nav2 for autonomous path planning and obstacle avoidance, cutting path-finding time by ~20% on competition courses

Stanford Pupper v3 Object Tracking — ROS2, Python, OpenCV

- Developed a real-time object tracking and following system on a quadruped robot, integrating custom ROS2 perception nodes with a neural locomotion controller
- Built an OpenCV-based detection pipeline that reliably identified and followed a target object, closing the loop between vision and physical motion

Visual Navigation Game — PyTorch, CNN, Python

- Built a vision-based navigation pipeline in PyTorch, tuning a CNN with SGD and regularization to improve goal-detection accuracy in a simulated robotic environment

Airline Booking System — Python, Flask, MySQL

- Designed a normalized relational schema across 8+ tables supporting a full booking lifecycle
- Implemented role-based authentication for customers and airline staff with session management and MD5 hashing

LEADERSHIP & PROFESSIONAL DEVELOPMENT

NYU Society of Women Engineers

Vice President

Jan 2026 – May 2026

New York, NY

- Led SWE's largest outreach initiative, launched a multi-tier mentorship program, and designed club newsletters

Finnegan, Henderson, Farabow, Garrett & Dunner LLP

Finnegan IP University Participant

Feb 2026

SKILLS

Languages & Frameworks: C++, Python, SQL, Flask, ROS2, PyTorch, OpenCV, Git, Gazebo, MATLAB

Hardware: Jetson, Raspberry Pi, Arduino, LiDAR